

straps are usually about 1.5m to 2.4m (5-feet to 8-feet) long, come with a standard alligator clip for ground ter-



Fig. 4: Belkin's anti-static wristband



Fig. 5: Zeebeetronics' wrist strap and dual footwear tester



Fig. 6: ESD tweezers



Fig. 7: ESD service box

mination and are the most-preferred over wireless wristbands for their reliability.

Other tools

ESD work-surface mat rolls, ESD floor-mat rolls, ESD cushion bags and boxes, ESD sheet protectors, ESD finger cots, ESD gloves, ESD jackets, ESD lables, ESD tapes and anti-static tweezers are some other items to consider depending on your application usage. There are a variety of anti-static bags available for storage and transportation of sensitive devices and components such as multilayer static shielding metallised bags, black conductive bags (usually seen with hard disks), pink bubble bags (seen in development boards packaging), foam covers and boxes.

Finger cots are available in different variants such as general-purpose finger cots, industrial-grade finger cots, nitrile finger cots, conductive finger cots and black or static dissipative finger cots. These are usually powder-free, made of latex material and are perfect for electronic assembly and manufacturing. Table mats used on work surfaces need to be grounded securely. Static-control test stations are also available to determine if ESD wrist straps or shoes are functioning properly.

In some situations, wearing wrist straps might not be viable. In such cases, use of ESD shoes or heel straps is recommended. In anti-static shoes, the grounding cord usually runs into the socks to make contact with the skin. ESD footwear are only effective with anti-static floor tiles.

Even basic equipment such as chairs, table, stools, cabinets, brushes and storage bins are available to create complete and secure anti-static workstations in facilities. In cases where air guns are required, one must

General selection criteria for ESD tools

- Buy products from trusted and reputed vendors of the industry
- Ensure good after-sales service and support is available
- Look for certified tools
- Ensure spare parts are available
- Warranty

To understand static electricity and its causes, triboelectric series, common sources of static charge and ESD sensitivity classifications, please refer to the Buyers' Guide featured on page 68 of April 2014 edition of Electronics For You. You may also visit www.electronicsforu.com and find this article under Buyers' Guide section.

make sure the guns do not generate static charges in the air stream. Special guns are available for this purpose.

Good-quality gloves provide anti-static protection up to 10^6 ohms and are nylon or polyester knit with conductive carbon fibres. ESD mats are available in single- and dual-layer forms. For general purpose, single-layer mats are preferred as these are cost-effective. ESD vinyl mats are ideal for static dissipative and flexible work surfaces. ESD tweezers are very handy and great to have if you are working with surface-mount or compact components.

Keep your devices and components safe

ESD can be a serious cause of malfunctions and crashes in today's sophisticated as well as simple electronic components and systems. With the best anti-static precautions, you can make sure your devices and components are ESD-protected, particularly when these are disassembled.

Warranty is an important factor to consider while buying ESD tools. Manufacturers typically offer 12 to 24 months of warranty for products such as ionisers, surface-resistivity meters, hot-air guns and wrist-strap testers. ●